

106/26 EXP NO.: 4 VERIFICATION OF THEVENIN'S THEOREM

AIM:

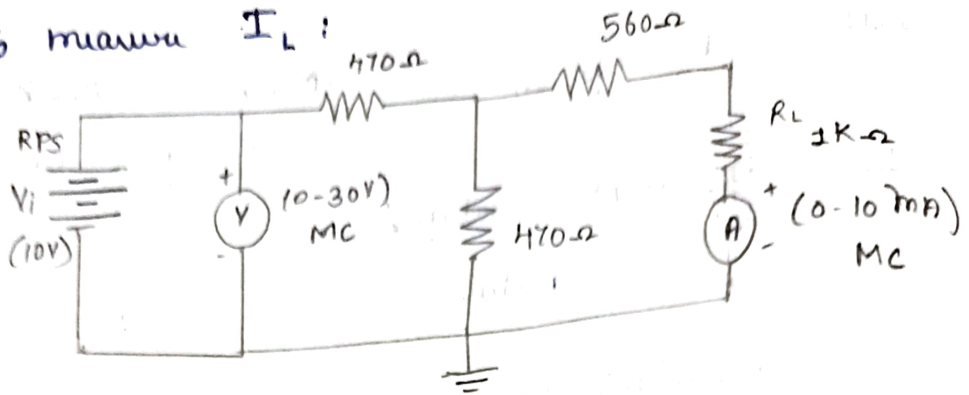
To verify Thevenin's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED:

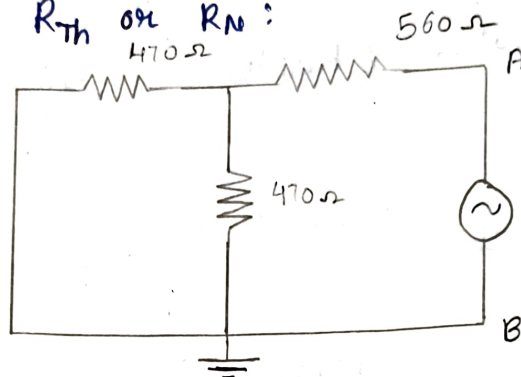
SL. No.	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated power supply	(0-30V)	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Resistors	470 Ω , 560 Ω , 1k Ω	2, 1, 1
5.	Bread board	—	1
6.	Multimeter	—	1

CIRCUIT DIAGRAM:

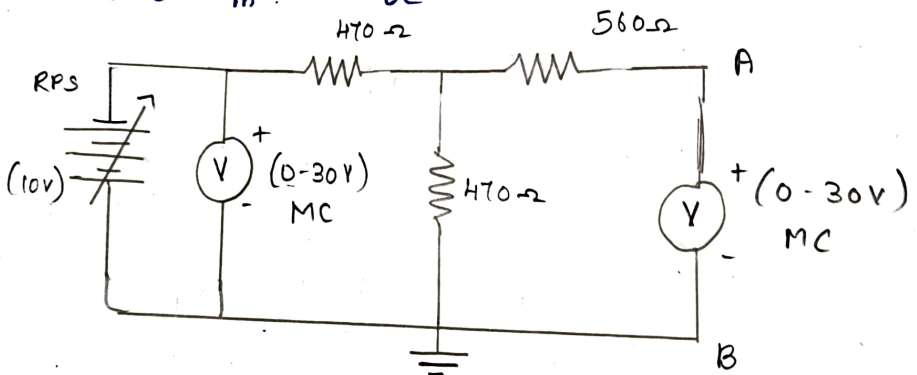
To measure I_L :



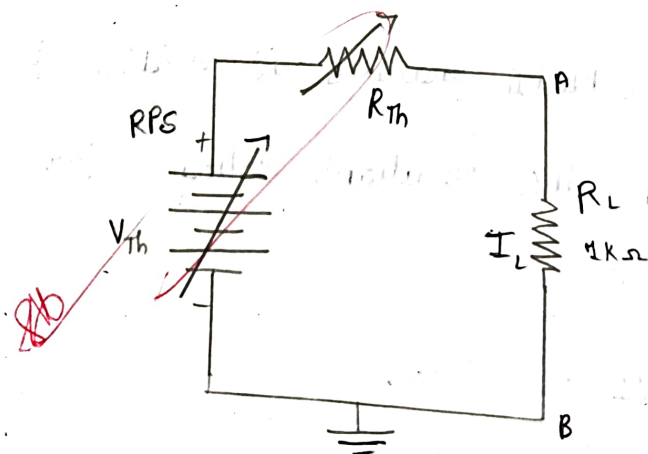
To measure R_{Th} or R_N :



To measure V_{Th} or V_{OC} :



Thermin's equivalent circuit:



PROCEDURE :

1. Make the connections as per the circuit diagram 1.
2. Vary the RPS and Set an input voltage of 10V.
3. Note down the voltmeter reading (V_i) and ammeter reading (I_L) in tabular column 1.
4. Switch off the Supply and make connections for circuit diagram 2.
5. Measure the Thevenin's resistance R_{Th} = Norton's resistance R_N .
6. Switch off the supply and make connections for circuit diagram 3.
7. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and $V_{Th} (=V_o)$ in tabular column : 3.
8. Switch off the Supply and make connections for circuit diagram 4.
9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and ammeter reading $I_N (=I_{sc})$ in tabular column 4.

TABULAR COLUMN : 1

To measure I_L

V_i (volts)	I_L (amps)
10V	2.8mA
8V	1.08mA
2V	0.36mA

TABULAR COLUMN : 2

To measure R_{Th} or R_N

From circuit diagram 2, $R_{Th} = R_N = 798\Omega$

TABULAR COLUMN : 3

V_i (volts)	V_{Th} (volts)
10V	5.07V
8V	2.60V
2V	0.65V

all verified

MODEL CALCULATIONS :

Practical Value of I_L (from tabulation I) = 2.3 mA

Verification of Thevenin's theorem

$$I_L = V_{Th} / (R_{Th} + R_L) = 2.22 \text{ mA}$$

Theoretical calculation of I_L , R_{Th} (R_N) and

V_{Th} for the given circuit :

THEORETICAL CALCULATIONS :

By voltage division rule, $V_{Th} = V_S \times \frac{R_3}{R_1 + R_3}$

$$V_{Th} = 10\text{V} \times \frac{470}{(470 + 470)} = 5\text{V}$$

$$V_{Th} = 5\text{V}$$

R_{Th} : R_1 & R_3 in parallel when voltage source in short circuited.

$$R_{Th} = \frac{470 \times 470}{470 + 470} + 560 = 235 + 560$$

$$R_{Th} = 795 \Omega$$

I_h :

R_{Th} in series with R_L

10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagrams 5 & 6 respectively.

11. Calculated the I_L value using the formula
Thevenin's theorem

$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's Theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

Theoretically verify the Norton's theorem.

$$I_L = \frac{V_{Th}}{R_{eq}} = \frac{5V}{1.795 \times 10^3 \Omega} = 2.79 \text{ mA}$$

$$I_L = 2.79 \text{ mA}$$

MARKS SPLIT-UP :

Sl. No.	Description	Marks allotted	Marks obtained
1.	Experimental Setup / Connections	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20

Total = 100

100

100
10/13

RESULT :

Thus, ~~Th~~venin's theorem is verified practically and theoretically.